

Design and Evaluation of Memristor-Based Piccolo-80 Lightweight Encryption Algorithm for Future IoT Applications

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Research Article

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Design and Evaluation of Memristor-Based Piccolo-80 Lightweight Encryption Algorithm for Future IoT Applications

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Abstract—Piccolo is an ultra-lightweight encryption algorithm optimized for very resource-constrained platforms. However, lightweight algorithms are vulnerable to side-channel attacks, specifically Differential Power Analysis (DPA). In this study, we first demonstrate the vulnerability of this algorithm against DPA. Then, a new masking scheme to protect the algorithm against DPA is proposed. In addition, the same implementation in a hybrid CMOS-Nano system is presented and the results are compared in terms of attack tolerance, area, and energy consumption. Power analysis mounted on unmasked Piccolo was successful while the attack on the masked version of both CMOS and CMOS-Nano implementations of the algorithm led to unsuccessful results and no key was recovered. More importantly, while maintaining reliable performance, the hybrid CMOS-Nano circuit was more efficient than the pure CMOS circuit in terms of energy consumption and area. This is a very important issue for future IoT devices and applications where energy consumption and security are the primary concerns.

Keywords: Nanoelectronics, Piccolo-80, Power Analysis Attacks, Countermeasure Techniques, CMOS-Nano circuits.

1. INTRODUCTION

The IoT has created new areas by connecting various devices to the network, but has also led to new security concerns. In 2020, the Information-Technology Promotion Agency of Japan (IPA) identified “Fraudulent Use of IoT Devices” as 9th major security threat in its annual report where almost 70% of IoT vendors have no concrete standards for their security responses in product development [1]. Lightweight cryptography is a solution developed to respond to security threats. However, these algorithms themselves are subject to differential power and electromagnetics analysis attacks [2, 3]. These attacks use power consumption or electromagnetic radiation profile of a crypto device to extract its sensitive information. Classical security solutions use algorithmic mechanisms that address a small subset of emerging security requirements, often at high energy. However, nanotechnology and nanoelectronics can change this situation. To improve the security and performance of digital integrated circuits other emerging and paradigms are needed. Piccolo is a hardware oriented cipher

algorithm which was presented in CHES’11 as an ultra-lightweight block cipher considering security, area and energy consumption as the most important design constraints. It is a 64-bit block cipher that supports 80-bits and 128-bit keys modes [4]. Nevertheless, this algorithm is vulnerable against differential power analysis. To the best of our knowledge, there is no credible report of this algorithm attacking or protecting against DPA. The only article we found in this regard is [5] in which the authors present a cipher text-based Correlation Power Analysis (CPA) targeting the last round of the algorithm. The simulation-based hypothetical power model presented in this paper is based on modeling the power consumption of D-type flip-flops in the last round of the algorithm. Their approach leads to $N \times 2^{20}$ correlation calculations to find 20-bits of the last round keys and takes at least two days to disclose the complete secret key, where N is the number of input plaintexts. In addition, it does not provide sufficient generality for various implementations on different platforms including microcontrollers or FPGAs.

In this paper, we discuss new solutions and contributions that nanotechnology may provide to the evolution of cryptographic hardware and embedded systems. We demonstrate how memristor as a nanoscale emerging device can be used for constructing security primitives in applications with very limited energy and resources [6]. At first, CPA was conducted on the whitening keys and round permutation of the first round of the algorithm. It was observed that the unprotected hardware implementation is vulnerable to CPA and effective countermeasures should be used. Then, the masking scheme proposed in [7] was used to mitigate the attack at the algorithm level. In addition, the same design was implemented using a hybrid CMOS-Memristor scheme and the results were compared in terms of attack tolerance, area and energy consumption.

Considering the common feature of memristor materials and the variation of their resistance with current which makes them a nonlinear multi-level element in nature compared with the CMOS transistor that is inherently a bi-level circuit element, power drawn from this nonlinear multi-level device with a pinched hysteresis loop is different to that of a Bi-state element [8]. This is a key feature in creating unpredictable power consumption to counter power analysis attacks at the technology and architecture level. Due to the special and unique characteristics of memristor as well as the compatibility of the manufacturing technologies, a lot of research is being done on mixed CMOS-Memristor circuits. It is now well-known that memristor is one of the leading and prominent emerging devices and technologies for advancing hardware security and computing systems [9, 10]. The novel nanoelectronics devices such as memristor can be harnessed for novel computing systems that are naturally immune to many existing conventional cyber-attacks. The CMOS-based scheme was implemented on Side-Channel Attack User Reference Architecture (SAKURA-G) board while the CMOS-Nano design was evaluated using Synopsys and Cadence Virtuoso IC6.1.4 tools along with a DPA attack software program written in MATLAB. The results demonstrated that hybrid CMOS-memristor circuit is more efficient than its CMOS counterpart. The area overhead to protect FPGA implementation of the algorithm against the attack is almost 15% while the hybrid CMOS-nano counterpart design requires only 160 more gate equivalent (GEs) to resist against power analysis attack. The approach presented in this article enables leveraging the beneficial aspects of both CMOS and nanotechnology. Briefly, this work includes the following novelties.

- It demonstrates the security vulnerability of the lightweight algorithm Piccolo against power analysis attack.
- It proposes an efficient masking scheme to mitigate the security weakness of the algorithm against power analysis attacks.
- It presents a new memristor-based design of the algorithm that is remarkably more efficient than its CMOS counterpart circuit in terms of energy consumption and utilized area.
- The results presented in this paper are important for platforms with very constrained resources such as future IoT applications in which energy consumption and security are the primary concerns.

In the rest of the paper, section 2 defines the position of nanoelectronics in the security pyramid of the future. The structure of cipher Piccolo is briefly reviewed in section 3. In section 4, the power analysis attacks are briefly illustrated. Fundamentals of the proposed Boolean masking scheme is presented in section 5. The results of power analysis attack on both CMOS-based and hybrid CMOS/memristor-based Piccolo are investigated in section 6. Finally, the results and discussions are summarized in the conclusions.

2. THE IMPORTANCE OF NANO SYSTEMS IN THE SECURITY OF FUTURE SYSTEMS

The field of electronic circuits and systems stands at the verge of a revolution driven by recent breakthroughs in emerging nanoscale devices. Especially in the field of computer security, there are signs that nanoelectronics can revolutionize this field in a big way. It is now well known that all existing electronic devices are vulnerable to various kinds of hardware and cyber-attacks. Among all the devices introduced for the transition from microelectronics to nanoelectronics, memristors, spintronics and carbon nanotubes are the most promising candidates. Their physical properties such as non-linearity, hysteresis, energy efficiency, integratability and CMOS compatibility make them very suitable candidates for new generation of circuits and security systems. To this end, future generation security projects should be organized in such a way that multiple areas and multiple interdisciplinary groups collaborate and work together. Fig. 1 shows different levels of the security pyramid and the position of nanoelectronics in it.

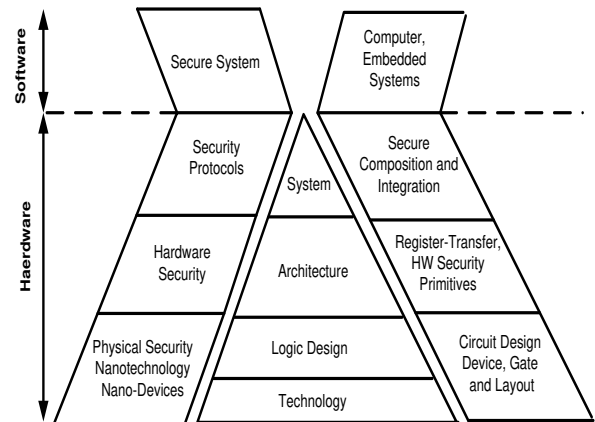


Fig. 1. Different levels of the security pyramid.

3. THE PICCOLO BLOCK CIPHER AND ITS IMPLEMENTATION

Piccolo is a lightweight 64-bit block encryption algorithm with an iterative structure that supports 80 and 128-bit keys, referred as Piccolo-80 and Piccolo-128, respectively. Both algorithms consist of a data processing part and a key scheduling part. The number of rounds for the data processing part and the key scheduling part are different in these two versions. It consists of a variant of generalized Feistel structure with four 16-bit branches and a permutation-based key schedule. The round function structure of the algorithm is very light and simple which consists of two nonlinear S-Box layers, separated by a diffusion matrix. S-Box of the algorithm is a non-linear substitution step and consists of four 4-bit bijective S-Boxes, as is shown in table I. It updates a 16-bit data $X_{(16)}$ as Eq. (1). The diffusion matrix M updates the 16-bit input data as is seen in Eq. (2).

Table I. Hexadecimal representation of 4-bit bijective S-Box S .

x	0	1	2	3	4	5	6	7	8	9	a	b	c	d	e	f
$S[x]$	e	4	b	2	3	8	0	9	1	A	7	f	6	c	5	d

$$(X_{0(4)}, X_{1(4)}, X_{2(4)}, X_{3(4)}) \leftarrow (S(X_{0(4)}), S(X_{1(4)}), S(X_{2(4)}), S(X_{3(4)})) \quad (1)$$

$$M = \begin{pmatrix} 2 & 3 & 1 & 1 \\ 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 1 & 1 & 2 \end{pmatrix}$$

$$(x_{0(4)}, x_{1(4)}, x_{2(4)}, x_{3(4)})^t \leftarrow M(x_{0(4)}, x_{1(4)}, x_{2(4)}, x_{3(4)})^t \quad (2)$$

The multiplications between matrix elements and 4-bit input vectors are carried out over $GF(2^4)$, constructed based on primitive polynomial $x^4 + x + 1$. The main structure of the algorithm consists of r rounds. It takes 64-bit plain data $X \in \{0, 1\}^{64}$, four 16-bit whitening keys, $wk_i \in \{0, 1\}^{16}$ ($0 \leq i < 4$) and $2r$ 16-bit round keys $rk_i \in \{0, 1\}^{16}$ ($0 \leq i < 2r$), as the inputs, and outputs a 64-bit encrypted data $Y \in \{0, 1\}^{64}$. The algorithm also divides 64-bit data in eight 8-bit groups and uses a word-based inter-round permutation to increase diffusion inside the cipher as much as possible. Fig. 2, Fig. 3 and Fig. 4 show the encryption function and its associated F-function, the structure of S-Box, and round permutation, respectively. As is seen, the S-Box of the algorithm is very lightweight and can be implemented using only four NORs, three XORs and one XNOR gate [4].

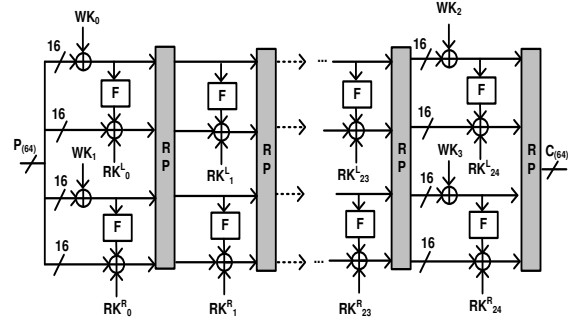


Fig. 2. Encryption function of Piccolo.

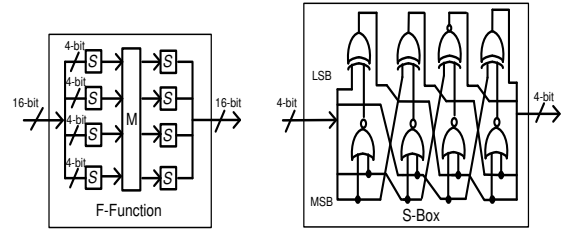


Fig. 3. (a) F-function and, (b) S-Box structure of Piccolo.

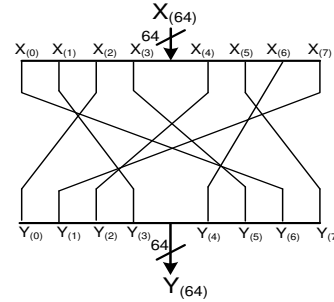


Fig. 4. The structure of Round Permutation RP .

4. Background of Power Analysis Attacks

Power analysis attacks are non-invasive hardware cryptanalytic attacks that use power consumption profile of the victim device to reveal its secret parameters or keys. This attack uses a basic fact that power consumption of an electronic device or chip is correlated to each active bit in the state of device at an instant in time and therefore leaks some information about it. In power analysis attack, an attacker uses a so-called hypothetical model of the attacked device, performs multiple power measurement and apply statistical methods to extract secret information. Ultimately, by comparing the correlation plots of power consumption for a particular algorithmic bit using power side-channel, one can find how a particular bit leaks in the device. Comparisons are made by applying statistical techniques on the data. Among the existing methods, the most common techniques are the distance-of-mean test and the correlation

power analysis. It has been shown that CPA is more efficient than DPA and correlation power analysis requires less power traces than the difference of means method.

In correlation power analysis, the model estimates the amount of information leakage for a specific moment of time in the execution. These estimations are correlated to the actual power measurements output. In a CPA attack, the correlation of the estimated Hamming distance changes, $h_{m,i}$ and the real power samples, $p_{m,i}$ are calculated using Pearson Correlation Coefficient. The results are used to compute the correlation coefficient matrix, $R_{i,j}$, as is seen in Eq.(3). Using this attack scenario, attacker determines and decides which subkey is most correlated to the real measured power traces [3].

$$R_{i,j} = \frac{\sum_{m=1}^M (P_{m,i} - \bar{P})(h_{m,i} - \bar{h})}{\sqrt{\sum_{m=1}^M (P_{m,i} - \bar{P})^2 \cdot \sum_{m=1}^M (h_{m,i} - \bar{h})^2}} \quad (3)$$

5. THE PROPOSED MODIFIED BOOLEAN MASKING SCHEME

The basic idea of masking is to mask intermediate key-dependent values of an encryption algorithm with a random number in order to eliminate the dependency between power consumption and sensitive parameters inside the encryption algorithm. Among the suggested first-order masking schemes, lookup-table masking and split masking are the most suggested and common methods. However, these techniques both suffer from a security vulnerability as they utilize the mask and the masked data at the same time. It has been also demonstrated that the split masking scheme is vulnerable to DPA regardless of the number of mask tables and of the possible modifications of the output masks. In-order to mask the S-Boxes of Piccolo, we used an improved and modified masking scheme in place of ordinary lookup-table or split masking techniques. This scheme stores randomly permuted masked S-box data in another table, S^* as is shown in Eq. (4)–Eq. (7) in such a way that actual S-Box values are not utilized or processed, where M stands for the pre-calculated mask table that is used to store the corresponding masked value for each table entry. n and m represent input and output masks, respectively [7].

$$S^*(p \oplus k \oplus n) = r(p \oplus k) \oplus S(p \oplus k \oplus n) + m \quad (4)$$

$$m(p \oplus k \oplus n) = S(p \oplus k \oplus n) \oplus S(p \oplus k) \quad (5)$$

$$M(p \oplus k \oplus n) = r(p \oplus k) \oplus S(p \oplus k \oplus n) \quad (6)$$

$$S(p \oplus k) \oplus m = S^*(p \oplus k \oplus n) \oplus M(p \oplus k \oplus n) \quad (7)$$

The corresponding procedure is shown as pseudocode in Algorithm I. At the start of encryption, each plaintext is added or masked with a random input number n , and then the corresponding mask table which each element is composed of the difference of actual SBox values and the output of the masked S-Box is computed and generated. In order to eliminate any dependency between actual values and power consumption of the device during fetch, instruction decode and execution of instructions, each element in S-Box table and its corresponding element in the mask table are added with a random value, r . This random value could be different for each table element. However, for the sake of simplicity, we used a constant value for all the table elements.

Algorithm I. The proposed masking method in pseudo-code for masking the Piccolo-80.

Require n, m

Output: $S(x) + m$

1. for $i = 0$ to 15 do
2. $S^*(i) = S(i) + r(i) + m$;
3. end
4. for $i = 0$ to 15 do
5. $M[(i + n) \bmod 15] = S(i) + S(i + n) + r(i)$;
6. end
7. $S(p + k) + m = M(p + k + n) + S^*(p + k + n)$
8. end

Return $S(x) + m$

6. Brief Review of Memristor

We used a modified version of Rak-Cserey model which is able to accurately characterize both current-voltage behavior of the memristor as well as the device dynamics and can be easily defined as a circuit element in the SPICE for our introductory simulations [11]. Fig. 5 shows the fundamental structure of the memristor. When a positive voltage is applied to the p-terminal (or a negative voltage is applied to the n-terminal) of the device, the oxygen vacancies, which form positive charges move towards the zero doped region. It increases the width w of the TiO_{2-x} layer. This results in the

device switching to a low resistance (R_{on}) state. However, when a negative voltage is applied to the p terminal (or a positive voltage is applied to the n-terminal) the oxygen vacancies move towards to the p-terminal. This results in the TiO_2 layer $D - w(t)$ expanding and TiO_{2-x} layer $w(t)$ decreasing. Hence, the device switches to a high resistance (R_{off}) state. It is obvious that when no voltage applied to either of two terminals, the boundary between the two titanium dioxide layers remains unchanged. This allows the memristor to maintain its previous state [8-11].

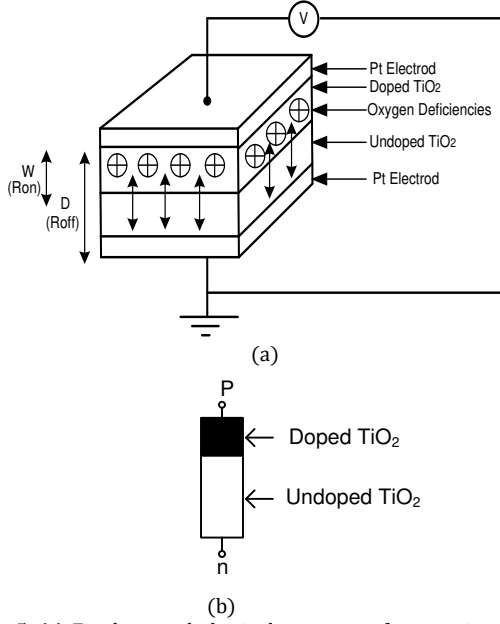


Fig. 5. (a) Fundamental physical structure of a memristor. The semiconductor film is placed between metal contacts, where w and D represent the thicknesses of the doped and total length of the memristor, respectively. (b) Symbol of the memristor.

After simulation of the basic modules in SPICE, the memristor was implemented using VTEAM-based Verilog-A codes. VTEAM is a general and accurate enough model that matches the target technology for practical memristive devices and circuits [12-14]. The memristor used has the following metrics: Model = 4, Window type = 0, $R_{on} = 50\Omega$, $dt = 1e-10$, $R_{off} = 10k\Omega$, $K_{on} = -30$ m/s, $K_{off} = 50$ m/s, $\alpha_{on} = 3$, $\alpha_{off} = 1$, $V_{on} = -0.2V$, $V_{off} = 0.02V$, where R_{on} , R_{off} , K_{on} , K_{off} , and α_{on} are the fitting parameters and V_{on} and V_{off} are SET and RESET voltage threshold. Due to the nature of the design, it was necessary to use a CAD tool with mixed signal simulation capability. Cadence IC Design is a very useful tool for simulation and evaluation of mixed signal designs. The memristor model was added to the library as a circuit component. Fig. 6 shows simulation result for current-voltage

hysteresis loop of the memristor in the Cadence IC Design environment driven by a voltage with an amplitude of 1.0 V and a frequency of 1 KHz. The individual components of the design were simulated and verified independently.

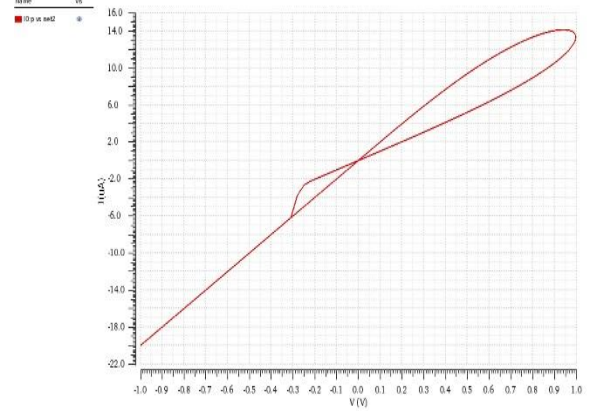


Fig. 6. Simulation result for current-voltage hysteresis loop of a memristor in the Cadence IC Design environment driven by a voltage with an amplitude of 1.0 V and frequency of 1 KHz.

7. FPGA IMPLEMENTATION OF THE ALGORITHM

Like most of the existing symmetric block ciphers, Piccolo includes two separate parts, the key scheduling part, which generates unique round keys and the data encryption part, which includes several similar rounds and utilize these keys. The key scheduling function for the 80-bit key mode, KS_r^{80} , divides the 80-bit key into five 16-bit separate subkeys $k_{i(16)}$ ($0 \leq i < 5$) and generates $wk_{i(16)}$ ($0 \leq i < 4$) and $rk_{j(16)}$ ($0 \leq j < 2r$) as is shown in Algorithm I, where K_i^L and K_i^R are left and right half 8 bits of k_i , respectively, i.e., $k_i(16) = K_i^L(8) \parallel K_i^R(8)$ and $K_i^R(8)$ contains the least significant bit of $k_i(16)$, c_i is a five-bit representation of i , e.g., $c_{12} = \{01100\}_2$. Algorithm II shows the encryption function, where F is a 16-bit F-function and RP is a 64-bit permutation defined in the section III. The number of rounds, r is 25 for Piccolo-80.

Algorithm II: Key Scheduling Algorithm $KS_r^{80}(K_{(80)})$
 Inputs: 80-bit Key: $K_{(80)} = K_0 \parallel K_1 \parallel K_2 \parallel K_3$
 Outputs: whitening keys: wk_{0-3} , round keys: rk_{0-2r}
 $wk_0 \leftarrow K_0^L \parallel K_1^R, wk_1 \leftarrow K_1^L \parallel K_0^R, wk_2 \leftarrow K_4^L \parallel K_3^R, wk_3 \leftarrow K_3^L \parallel K_4^R$
 for $i = 0 : r - 1$
 $(rk_{2i}, rk_{2i+1}) \leftarrow (con_{2i}^{80}, con_{2i+1}^{80})$
 $\oplus \begin{cases} (k_2, k_3) & (if\ i \bmod 5 = 0\ or\ 2) \\ (k_0, k_1) & (if\ i \bmod 5 = 1\ or\ 4) \\ (k_4, k_4) & (if\ i \bmod 5 = 3) \end{cases}$
 $(con_{2i}^{80} \parallel con_{2i+1}^{80}) \leftarrow (c_{i+1} \parallel c_0 \parallel c_{i+1} \parallel \{00\}_2 \parallel c_{i+1} \parallel c_0 \parallel c_{i+1})$
 $\oplus \{0f1e2d3c\}_{16}$

Algorithm III. Encryption Function

Inputs: Plaintexts $P_{(64)}, wk_{0-3}, rk_{0-(2r-1)}$

Outputs: Cipher Text $C_{(64)}$

$E_r(P_{(64)}, wk_{(0)}, \dots, wk_{(3)}, rk_{(0)}, \dots, rk_{(2r-1)})$

$P_{0(16)}|P_{1(16)}|P_{2(16)}|P_{3(16)} \leftarrow P_{(64)}$

$P_0 \leftarrow P_0 \oplus wk_0, P_2 \leftarrow P_2 \oplus wk_1$

for $i = 0: r - 2$

$P_1 \leftarrow P_1 \oplus F(P_0) \oplus rk_{2i}, P_3 \leftarrow P_3 \oplus F(P_2) \oplus rk_{2i+1}$

$P_0|P_1|P_2|P_3 \leftarrow RP(P_0|P_1|P_2|P_3)$

$P_1 \leftarrow P_1 \oplus F(P_0) \oplus rk_{2r-2}, P_3 \leftarrow P_3 \oplus F(P_2) \oplus rk_{2r-2}$

$P_0 \leftarrow P_0 \oplus wk_2, P_2 \leftarrow P_2 \oplus wk_3$

$C_{(64)} = P_0|P_1|P_2|P_3$

As mentioned, the only major arithmetic function in the structure of Piccolo is to multiply matrix elements M on 4-bit input vectors over $GF(2^4)$. It can be implemented using a serial or parallel structure or can be implemented over composite field $GF((2^2)^2)$. In this work we utilized a parallel $GF(2^4)$ multiplier in which the multiplication is accomplished in only one clock cycle. In general, a $GF(2^m)$ parallel multiplier requires $(m + 1)$, m -input modulo-2 adders, $m(m + 1)$ two-inputs AND gates and m two-inputs XOR gates. Fortunately, this structure is quite compatible with our design. In order to design a polynomial bases $GF(2^4)$ multiplier that multiplies two input vectors $P = (p_3, p_2, p_1, p_0)$ and $Q = (q_3, q_2, q_1, q_0)$, modulo $f(X) = X^4 + X + 1$, at first, the two inputs are multiplied normally. Then, the result is reduced and can be represented as follows:

$$\begin{aligned}
 P(x) \cdot Q(x) \bmod f(x) = & \\
 (p_0q_3 + p_1q_2 + p_2q_1 + p_3q_0 + p_3q_3)x^3 + & (p_0q_2 + p_1q_1 + p_2q_0 + p_2q_3 + p_3q_2 + p_3q_3)x^2 + \\
 (p_0q_1 + p_1q_0 + p_1q_3 + p_2q_2 + p_2q_3 + p_3q_1 + & p_3q_2)x + (p_0q_0 + p_1q_3 + p_2q_2 + p_3q_1) \quad (10)
 \end{aligned}$$

Eq. (10) can be simplified and rewritten as follows:

$$\begin{aligned}
 P(x) \cdot Q(x) \bmod f(x) = & t_3x^3 + t_2x^2 + t_1x + t_0 \quad (11)
 \end{aligned}$$

$$\begin{aligned}
 t_0 = & \overline{(a_0b_0 \oplus 1)} \oplus \overline{(a_1b_3 \oplus 1)} \\
 & \oplus \overline{(a_2b_2 \oplus 1)} \oplus \overline{(a_3b_1 \oplus 1)} \\
 t_1 = & \overline{\overline{(a_0b_1 \oplus 1)} \oplus \overline{(a_1b_0 \oplus 1)} \oplus \overline{(a_1b_3 \oplus 1)} \oplus \overline{(a_2b_2 \oplus 1)}} \\
 & \oplus \overline{(a_2b_3 \oplus 1)} \oplus \overline{(a_3b_1 \oplus 1)} \oplus \overline{(a_3b_2 \oplus 1)} \\
 t_2 = & \overline{\overline{(a_0b_0 \oplus 1)} \oplus \overline{(a_0b_0 \oplus 1)} \oplus \overline{(a_0b_0 \oplus 1)} \oplus \overline{(a_0b_0 \oplus 1)}} \\
 & \oplus \overline{(a_0b_0 \oplus 1)} \oplus \overline{(a_0b_0 \oplus 1)} \\
 t_3 = & \overline{\overline{(a_0b_3 \oplus 1)} \oplus \overline{(a_1b_2 \oplus 1)} \oplus \overline{(a_2b_1 \oplus 1)} \oplus \overline{(a_3b_0 \oplus 1)}} \oplus \\
 & \overline{(a_3b_3 \oplus 1)} \quad (12)
 \end{aligned}$$

Fig. 7 shows the implementation of the parallel $GF(2^4)$ multiplier using Boolean gates. In general, a parallel $GF(2^m)$ architecture requires $(m + 1)$ m -input mod 2 adders, $m(m + 1)$ two-input AND gates, and m two-input XOR gates. The m two-input XOR gates are employed to either complement or not complement the outputs of the top-most m mod 2 adders depending upon the output of the bottom m -input mod 2 adder. In summary, this structure can be realized using 16 XOR and 16 NAND gates with 4 XOR and 1 NAND in the critical path or 18 XOR and 12 AND with 4 XOR and 1 AND in the critical path.

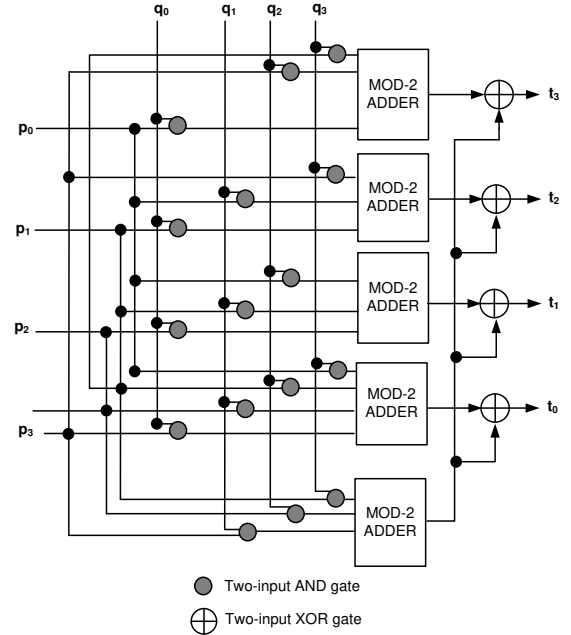


Fig. 7. Implementation of parallel $GF(2^4)$ multiplier using mod-2 adders and Boolean gates.

Since area and energy are major concerns for the applications that use lightweight cryptography like IoT applications, we optimized our design for area. As we know, an architecture that targets area is one that reuses the logic resources as much as possible, often at the cost of throughput or speed. In most cases this requires a recursive data path, where the output of the main stage is fed back to the input for the same processing. Such architectures can use a simple loop that flows naturally with the algorithm or they may require special controls for more complex logic reuse. As mentioned, Piccolo is a variant of generalized Feistel network with a permutation based key schedule. The structure of all rounds are the same except that the last round does not contain round permutation. In this design, we did not use LUT-based Galois Field multiplier for $GF(2^4)$ multiplication and used the logic-based

implementation as was shown in Fig.8. However, we used a simple LUT-based SBox for the implementation of the non-linear function since the SBox structure of the algorithm is very simple. Fig. 9 shows the block diagram of the designed hardware architecture for the algorithm. It should also be noted that, Round keys can either be provided beforehand and stored in memory or be generated on the fly. In the applications with limited area, generating round keys on the fly is a better choice.

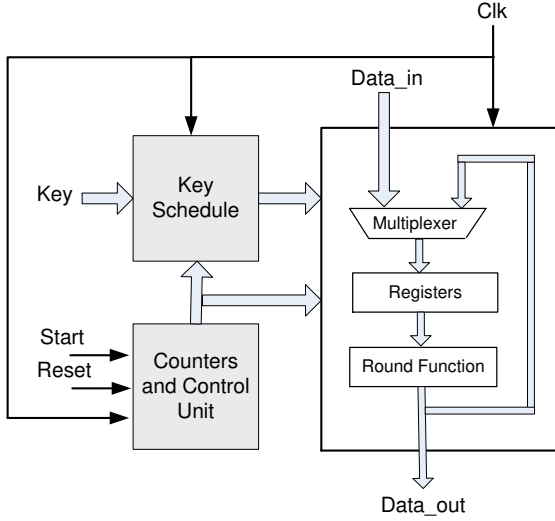


Fig. 8. The basic architecture of the implementation of the algorithm in this work.

8. CMOS-MEMRISTOR ARCHITECTURE FOR PICCOLO-80

As mentioned, the whole Piccolo-80 can be implemented using basic Boolean gates such as XOR and NAND gates. Fig. 9 shows the basic architecture for the memristor-based XOR circuit which is very suitable for our purpose [15, 16]. It consists of four memristors M_1 , M_2 , M_3 and M_4 . Here, M_1 and M_3 are connected to form basic AND operation. M_2 and M_4 are connected to form basic OR operation. The voltage difference between V_{L1} and V_{L2} creates XOR operation in a single clock cycle as is shown in Table II. In Table II, the logical operation of the XOR circuit that only uses the memristor is shown. The operation of the circuit is based on whether or not current flows from V_{L1} to V_{L2} through the load resistance R_L .

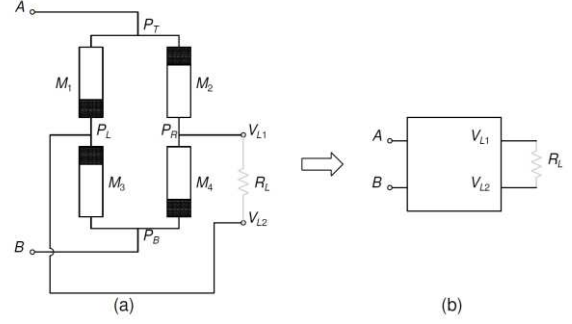


Fig. 9. Memristive XOR circuit. (a) Purely Memristive XOR circuit, (b) Symbol used for the circuit [15].

Table II. Performance of the Memristive XOR function.

Inputs		$V_{L1}-V_{L2}$	Output
$A = 0$	$B = 0$	0	0
$A = V_{cc}$	$B = 0$	V_{cc}	V_{cc}
$A = 0$	$B = V_{cc}$	V_{cc}	V_{cc}
$A = V_{cc}$	$B = V_{cc}$	0	0

Fig. 10 shows the buffered Memristive XOR/AND and XNOR/OR gates with dual functionality. Fig. 11 shows the full-adder circuit based on the same multifunction logic blocks. The sum and the output carry can be easily created using the following equations and the same circuits.

$$S = (A \oplus B) \oplus 1 \oplus C_i \oplus 1 = A \oplus B \oplus C_i \quad (8)$$

$$\begin{aligned} C_o &= C_i(A \oplus B) + AB = C_i(\bar{A}\bar{B} + \bar{B}\bar{A}) + AB \\ &= \bar{A}\bar{B}C_i + \bar{A}B\bar{C}_i + A\bar{B}C_i + AB = (AB + \bar{A}\bar{B})(AB + AC) + \bar{A}\bar{B}C_i \\ &= AB + AC + \bar{A}\bar{B}C_i = AB + C(A + \bar{A}) \\ &= AB + C(A + \bar{A})(A + B) = AB + AC + BC \end{aligned} \quad (9)$$

At first, we simulated the XOR and NAND gates, which are the basic gates in the algorithm implementation, as well as the proposed full-adder circuit in Cadence environment to verify the correctness of these topologies. The results are shown in Fig. 12 and Fig. 13, respectively. Fig. 14 and Fig. 15 show the transient response and power consumption of a CMOS-memristor NAND gate using the same logic style in the same environment, respectively. Ultimately, the whole algorithm was implemented in an unrolled architecture. Using a python script, the input data and their associated power traces were stored, paired and exported to MATLAB for the required subsequent data and power attack processing.

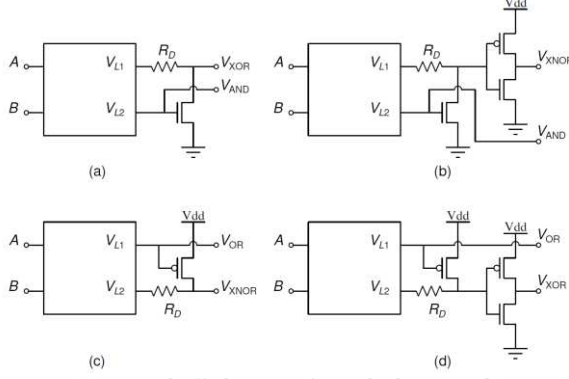


Fig. 10. (a) 1T-4M bufferless XOR/AND dual circuit, (b) 3T-4M fully buffered XNOR circuit, (c) 1T-4M bufferless XNOR/OR dual functionality circuit and (d) 3T-4M fully buffered XOR circuit [13].

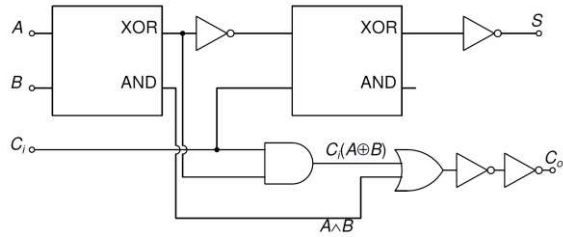


Fig. 11. Buffered hybrid CMOS-Memristor full Adder circuit [15].

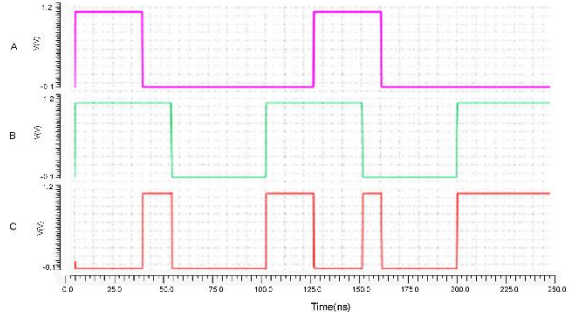


Fig. 12. Transient response of the CMOS-Memristor based XOR gate to the inputs in the Cadence IC Design environment.

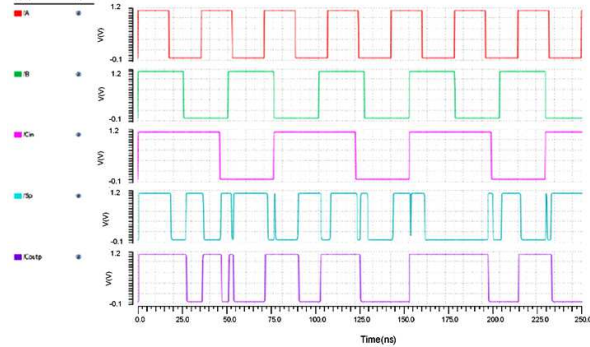


Fig. 13. Simulation of the full adder in Cadence IC design environment. Top three signals represent the inputs A, B and input carry, C_i , respectively, and the bottom two signals represent the sum (S) and output carry (C_o), respectively.

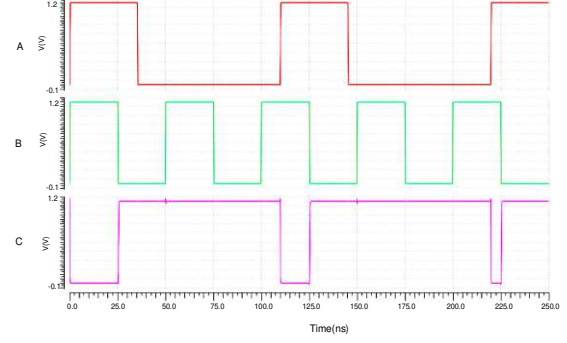


Fig. 14. Transient response of the CMOS-Memristor based NAND gate to the inputs in the Cadence IC Design environment.

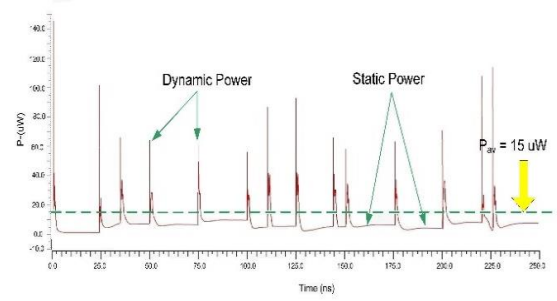


Fig. 15. Power consumption of a 2-input NAND gate.

9. CORRELATION POWER ATTACK TO PICCOLO-80

As was described, 16-bit input plaintext is XORed with the 16-bit whitening keys wk_0 and wk_1 and then enters the F-Function. Like most of the other block ciphers, Piccolo suffers from security vulnerability against DPA in its non-linear layer. A 4-bit selection function can be used to partition traces and assign them to corresponding subsets. Then, the output of the first substitution layer of the first round function of the algorithm can be targeted to recover wk_0 and wk_1 . Round keys rk_0 and rk_1 could be recovered by applying the same procedure in the second round of the algorithm. We used a Hamming-Distance based CPA to recover the secret keys.

In order to evaluate the robustness of the both design schemes, the algorithm in 25-round mode was implemented on Side-Channel Attack User Reference Architecture (SAKURA-G) board which includes a Xilinx Spartan-6 XC6SLX9 FPGA and some control circuits. The power traces were recorded by Infiniium Keysight DSO90604A digital oscilloscope with 20 GS/sec and BW=6 GHz, and an Agilent SN MY 6596/4PEA current probe to measure the board's supply current. Fig. 16 shows a photo of the SAKURA board used in this project. Applying 1000 plaintexts to the input yields a successful CPA where the correct subkey is clearly revealed while the same experiment on the

protected implementation is unsuccessful as is shown in Fig. 17.



Fig. 16. Photograph of SAKURA-G board which was used in our experiments.

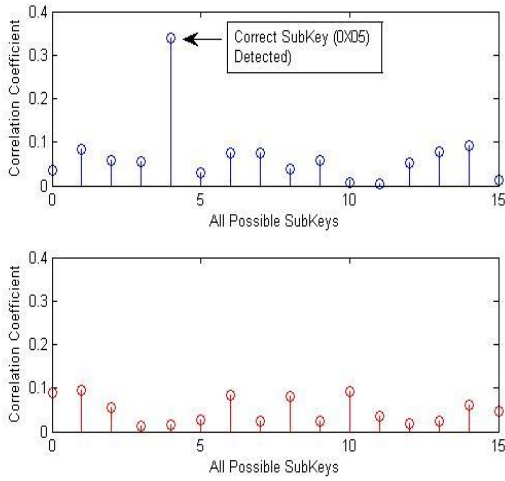


Fig. 17. Result of CPA attack to $wk_0(0:4)$ in the FPGA implementation after applying 1000 plaintext. Upper figure: unmasked implementation. Lower figure: masked implementation.

We also performed Test Vector Leakage Assessment (TVLA) or t-test that is a classical and extensively used methodology for side-channel leakage detection and assessment. To perform a specific t-test, 3000 plaintexts were applied to the algorithm and the power traces were divided into two sets in accordance with the hypothetical values of the S-Box output. The two trace sets were considered as two normally distributed populations. Then, Welch's statistical t-test was applied by testing the difference of the two population means over each time instant in the traces as is seen in Eq. (13), where, T_A , T_B , S_A , S_B , T_A , and N_B represent the means, variances, and size of the two trace sets, respectively [18].

$$t = \frac{(\overline{T_A} - \overline{T_B})}{\sqrt{\left(\frac{S_A^2}{N_A}\right) + \left(\frac{S_B^2}{N_B}\right)}} \quad (13)$$

The result of T-test for the FPGA implementation is shown in Fig. 18. The result of CPA to hybrid CMOS-Nano circuit after applying 1500 plaintexts is shown in Fig. 19. Table III compares the two implementations in the terms of area and average power consumption. Both circuits were implemented in round-based mode in which one round is processed within one clock cycle. CMOS 65 nm technology at the standard 1.2V was used. In this table, GE stands for gate equivalent. One GE is equivalent a 2-input AND gate area. The average power consumed in the CMOS-memristor circuit is 22.5 mW which was calculated at frequency of 120 MHz which is close to the maximum possible operating frequency of the hybrid circuit. The power measurements are based on the sum of the static and dynamic powers reported by Cadence. The values of R_{ON} and R_{OFF} of the memristor are key factors in power consumption of the circuit while memristor with high values of R_{OFF} is under research and development. In order to decide how efficient a design is, we utilize the efficiency defined as a figure of merit.

$$\frac{\text{Throughput}}{\text{Area}} \left(\frac{\text{Mbit}}{\text{s}} \right) \left(\frac{\text{Slices}}{\text{Slices}} \right)$$

where throughput is defined as:

$$\frac{\text{Working frequency} \times \text{Number of Bits}}{\text{Number of Clock Cycles to process one block}}$$

It should also be noted that there are numerous CMOS implementations of encryption algorithms in the open literature, however, there are few reports on Piccolo implementation for comparison. In addition, we chose the same or a close target platform as the SAKURA board to have a fair comparison with other works.

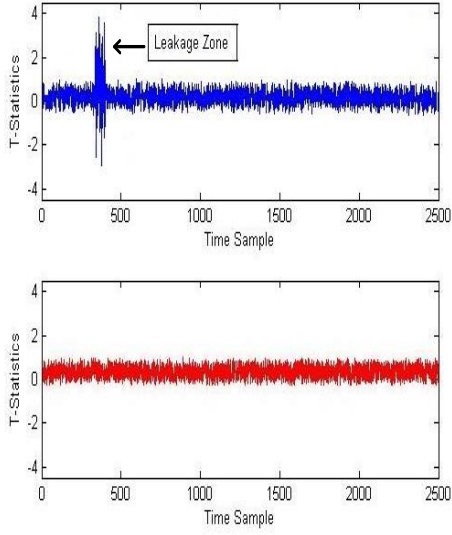


Fig. 18. Detection results of Welch's t-test after applying 3000 plaintexts to the FPGA implementation. The upper figure: the unmasked implementation, the lower figure: the masked implementation.

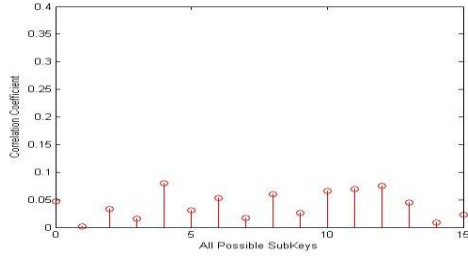


Fig. 19. Result of CPA attack to $wk_0(0:4)$ in the mixed CMOS-Nano circuit after applying 1500 plaintext.

Table III. Implementation results for Piccolo-80.

Algorithm/Reference	Device	Slices/GEs	LUTs	Freq. (MHz)	Throughput	Power Consumption (mW)
Piccolo-80 [19]	Spartan-6 xc6slx25	184	563	189	1.02	191
Piccolo-80 [19]	xc4vlx25	273	525	273	2.45	148
Piccolo-80 [20]	xc5vlx50t-3ff1136	47	150	315	7.81	560
Piccolo-80 [20]	Viertex-4 xc4vlx25	111	209	228	2.39	273
This Work (CMOS-Unprotected)	Spartan-6 xc6slx9	64	126	206	3.2	86
This Work (CMOS-Protected)	Spartan-6 xc6slx9	73	138	188	2.3	105
This Work (Unprotected-Hybrid MeMOS)	-	1352	-	120	-	22.5
This Work (Protected- Hybrid MeMOS)	-	1512	-	112	-	25.6

10. CONCLUSIONS

In this work, the weakness of ultra-compact Piccolo-80 against power analysis was demonstrated. Also, an efficient and low cost masking technique was proposed to overcome this vulnerability. The area overhead to protect FPGA implementation of the algorithm against the attack is almost 15% while the hybrid CMOS-Nano counterpart design requires only 160 more GEs to resist against power analysis attack. Almost all of the existing security protection mechanisms can be defeated and none of the existing techniques provides perfect security, instead they increase the number of required side-channel measurements, and hence, hardly improve the resistance against side-channel attacks. However, to countermeasure side-channel and tampering attacks other emerging and paradigms are needed. Nanoscale emerging electronic devices, once again, could offer the possibility of addressing a lot of security shortcomings. There are many solutions and ideas in the nanotechnology and perhaps other communities to resolve existing security issues such as counterfeiting, illicit copying of products and intellectual properties. One potential solution is to use variations in resistive states of memristors in various architectures to fabricate on-chip memristive device-based security primitives. However, fabricating high-quality materials, overcoming reliability issues, developing improved device models and powerful CAD tools remain challenging. In addition, application of other emerging devices such as spintronic, carbon nanotube and transistors ... in hardware security including countermeasure to counterfeiting, hardware Trojans, unauthorized access or modification of data and physical attacks and microprobing needs to be further investigated and explored [5], [21-25].

Declarations

Ethical Approval

Not Applicable

Competing Interests

There is no competing interests in this study.

Authors' contributions

Not Applicable

Funding

There is no funding in this study.

Availability of data and materials

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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